Assignment 4 - FMLA

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a. Use only the numerical variables (1 to 9) to cluster the 21 firms.

Load the Dataset

```
Pharma <- read.csv("~/ML/Pharmaceuticals.csv")

head(Pharma)
```

```
##   Symbol                Name Market_Cap Beta PE_Ratio  ROE  ROA Asset_Turnover
## 1    ABT Abbott Laboratories      68.44 0.32     24.7 26.4 11.8            0.7
## 2    AGN      Allergan, Inc.       7.58 0.41     82.5 12.9  5.5            0.9
## 3    AHM        Amersham plc       6.30 0.46     20.7 14.9  7.8            0.9
## 4    AZN     AstraZeneca PLC      67.63 0.52     21.5 27.4 15.4            0.9
## 5    AVE             Aventis      47.16 0.32     20.1 21.8  7.5            0.6
## 6    BAY           Bayer AG       16.90 1.11     27.9  3.9  1.4            0.6
##   Leverage Rev_Growth Net_Profit_Margin Median_Recommendation Location Exchange
## 1     0.42       7.54              16.1          Moderate Buy       US     NYSE
## 2     0.60       9.16               5.5          Moderate Buy   CANADA     NYSE
## 3     0.27       7.05              11.2            Strong Buy       UK     NYSE
## 4     0.00      15.00              18.0         Moderate Sell       UK     NYSE
## 5     0.34      26.81              12.9          Moderate Buy   FRANCE     NYSE
## 6     0.00      -3.17               2.6                  Hold  GERMANY     NYSE
```

```
str(Pharma)
```

```
## 'data.frame':    21 obs. of  14 variables:
##  $ Symbol               : chr  "ABT" "AGN" "AHM" "AZN" ...
##  $ Name                 : chr  "Abbott Laboratories" "Allergan, Inc." "Amersham plc" "AstraZeneca PL
##  $ Market_Cap           : num  68.44 7.58 6.3 67.63 47.16 ...
##  $ Beta                 : num  0.32 0.41 0.46 0.52 0.32 1.11 0.5 0.85 1.08 0.18 ...
##  $ PE_Ratio             : num  24.7 82.5 20.7 21.5 20.1 27.9 13.9 26 3.6 27.9 ...
##  $ ROE                  : num  26.4 12.9 14.9 27.4 21.8 3.9 34.8 24.1 15.1 31 ...
##  $ ROA                  : num  11.8 5.5 7.8 15.4 7.5 1.4 15.1 4.3 5.1 13.5 ...
##  $ Asset_Turnover       : num  0.7 0.9 0.9 0.9 0.6 0.6 0.9 0.6 0.3 0.6 ...
##  $ Leverage             : num  0.42 0.6 0.27 0 0.34 0 0.57 3.51 1.07 0.53 ...
##  $ Rev_Growth           : num  7.54 9.16 7.05 15 26.81 ...
##  $ Net_Profit_Margin    : num  16.1 5.5 11.2 18 12.9 2.6 20.6 7.5 13.3 23.4 ...
##  $ Median_Recommendation: chr  "Moderate Buy" "Moderate Buy" "Strong Buy" "Moderate Sell" ...
```

```
##  $ Location             : chr  "US" "CANADA" "UK" "UK" ...
##  $ Exchange             : chr  "NYSE" "NYSE" "NYSE" "NYSE" ...
```

```
summary(Pharma)
```

```
##        Symbol           Name      Market_Cap          Beta
##  Length   :21   Length   :21   Min.   :  0.41   Min.   :0.1800
##  N.unique :21   N.unique :21   1st Qu.:  6.30   1st Qu.:0.3500
##  N.blank  : 0   N.blank  : 0   Median : 48.19   Median :0.4600
##  Min.nchar: 3   Min.nchar: 5   Mean   : 57.65   Mean   :0.5257
##  Max.nchar: 4   Max.nchar:34   3rd Qu.: 73.84   3rd Qu.:0.6500
##                                Max.   :199.47   Max.   :1.1100
##     PE_Ratio          ROE            ROA        Asset_Turnover    Leverage
##  Min.   : 3.60   Min.   : 3.9   Min.   : 1.40   Min.   :0.3    Min.   :0.0000
##  1st Qu.:18.90   1st Qu.:14.9   1st Qu.: 5.70   1st Qu.:0.6    1st Qu.:0.1600
##  Median :21.50   Median :22.6   Median :11.20   Median :0.6    Median :0.3400
##  Mean   :25.46   Mean   :25.8   Mean   :10.51   Mean   :0.7    Mean   :0.5857
##  3rd Qu.:27.90   3rd Qu.:31.0   3rd Qu.:15.00   3rd Qu.:0.9    3rd Qu.:0.6000
##  Max.   :82.50   Max.   :62.9   Max.   :20.30   Max.   :1.1    Max.   :3.5100
##    Rev_Growth    Net_Profit_Margin Median_Recommendation      Location
##  Min.   :-3.17   Min.   : 2.6      Length   :21           Length   :21
##  1st Qu.: 6.38   1st Qu.:11.2      N.unique : 4           N.unique : 7
##  Median : 9.37   Median :16.1      N.blank  : 0           N.blank  : 0
##  Mean   :13.37   Mean   :15.7      Min.nchar: 4           Min.nchar: 2
##  3rd Qu.:21.87   3rd Qu.:21.1      Max.nchar:13           Max.nchar:11
##  Max.   :34.21   Max.   :25.5
##       Exchange
##  Length   :21
##  N.unique : 3
##  N.blank  : 0
##  Min.nchar: 4
##  Max.nchar: 6
##
```

```
dim(Pharma)
```

```
## [1] 21 14
```

Extract the numerical variables

```
Pharma_numeric <- Pharma[, 3:11]

# View selected variables

head(Pharma_numeric)
```

```
##   Market_Cap Beta PE_Ratio  ROE  ROA Asset_Turnover Leverage Rev_Growth
## 1      68.44 0.32     24.7 26.4 11.8            0.7     0.42       7.54
## 2       7.58 0.41     82.5 12.9  5.5            0.9     0.60       9.16
```

```
## 3       6.30 0.46     20.7 14.9  7.8            0.9     0.27       7.05
## 4      67.63 0.52     21.5 27.4 15.4            0.9     0.00      15.00
## 5      47.16 0.32     20.1 21.8  7.5            0.6     0.34      26.81
## 6      16.90 1.11     27.9  3.9  1.4            0.6     0.00      -3.17
##   Net_Profit_Margin
## 1              16.1
## 2               5.5
## 3              11.2
## 4              18.0
## 5              12.9
## 6               2.6
```

#Check for missing values

```
sum(is.na(Pharma_numeric))
```

```
## [1] 0
```

#Scale the variables

```
# Standardize numerical variables
Pharma_scaled <- scale(Pharma_numeric)

# View standardized data
head(Pharma_scaled)
```

```
##      Market_Cap        Beta    PE_Ratio         ROE        ROA Asset_Turnover
## [1,]  0.1840960 -0.80125356 -0.04671323  0.04009035  0.2416121      0.0000000
## [2,] -0.8544181 -0.45070513  3.49706911 -0.85483986 -0.9422871      0.9225312
## [3,] -0.8762600 -0.25595600 -0.29195768 -0.72225761 -0.5100700      0.9225312
## [4,]  0.1702742 -0.02225704 -0.24290879  0.10638147  0.9181259      0.9225312
## [5,] -0.1790256 -0.80125356 -0.32874435 -0.26484883 -0.5664461     -0.4612656
## [6,] -0.6953818  2.27578267  0.14948233 -1.45146000 -1.7127612     -0.4612656
##        Leverage Rev_Growth Net_Profit_Margin
## [1,] -0.2120979 -0.5277675        0.06168225
## [2,]  0.0182843 -0.3811391       -1.55366706
## [3,] -0.4040831 -0.5721181       -0.68503583
## [4,] -0.7496565  0.1474473        0.35122600
## [5,] -0.3144900  1.2163867       -0.42597037
## [6,] -0.7496565 -1.4971443       -1.99560225
```

#Run K-Means for k=2,3 and 4

```
set.seed(123)

k2 <- kmeans(Pharma_scaled,
             centers = 2,
             nstart = 25)

k3 <- kmeans(Pharma_scaled,
             centers = 3,
             nstart = 25)
```

```r
k4 <- kmeans(Pharma_scaled,
             centers = 4,
             nstart = 25)
```

Add Cluster labels to the Dataset

```r
Pharma$Cluster_k2 <- k2$cluster
Pharma$Cluster_k3 <- k3$cluster
Pharma$Cluster_k4 <- k4$cluster

# View company cluster assignments
head(Pharma)
```

```
##   Symbol                Name Market_Cap Beta PE_Ratio  ROE  ROA Asset_Turnover
## 1    ABT Abbott Laboratories      68.44 0.32     24.7 26.4 11.8            0.7
## 2    AGN      Allergan, Inc.       7.58 0.41     82.5 12.9  5.5            0.9
## 3    AHM        Amersham plc       6.30 0.46     20.7 14.9  7.8            0.9
## 4    AZN     AstraZeneca PLC      67.63 0.52     21.5 27.4 15.4            0.9
## 5    AVE             Aventis      47.16 0.32     20.1 21.8  7.5            0.6
## 6    BAY            Bayer AG      16.90 1.11     27.9  3.9  1.4            0.6
##   Leverage Rev_Growth Net_Profit_Margin Median_Recommendation Location Exchange
## 1     0.42       7.54              16.1          Moderate Buy       US     NYSE
## 2     0.60       9.16               5.5          Moderate Buy   CANADA     NYSE
## 3     0.27       7.05              11.2            Strong Buy       UK     NYSE
## 4     0.00      15.00              18.0         Moderate Sell       UK     NYSE
## 5     0.34      26.81              12.9          Moderate Buy   FRANCE     NYSE
## 6     0.00      -3.17               2.6                  Hold  GERMANY     NYSE
##   Cluster_k2 Cluster_k3 Cluster_k4
## 1          1          2          1
## 2          2          3          4
## 3          2          3          1
## 4          1          2          1
## 5          2          1          2
## 6          2          3          4
```

Compute Cluster Summaries for K = 2

```r
aggregate(Pharma_numeric,
          by = list(Cluster = k2$cluster),
          FUN = mean)
```

```
##   Cluster Market_Cap      Beta PE_Ratio  ROE      ROA Asset_Turnover  Leverage
## 1       1   97.11364 0.4336364 20.95455 35.7 14.95455           0.80 0.3254545
## 2       2   14.24300 0.6270000 30.42000 14.9  5.63000           0.59 0.8720000
##   Rev_Growth Net_Profit_Margin
## 1   10.16455          20.17273
## 2   16.89800          10.77000
```

For K = 3

Calculate average values

```
aggregate(Pharma_numeric,
          by = list(Cluster = k3$cluster),
          FUN = mean)
```

```
##   Cluster Market_Cap      Beta PE_Ratio  ROE       ROA Asset_Turnover  Leverage
## 1       1    9.23500 0.6483333 19.43333 17.3  5.983333      0.4833333 1.2500000
## 2       2   97.11364 0.4336364 20.95455 35.7 14.954545      0.8000000 0.3254545
## 3       3   21.75500 0.5950000 46.90000 11.3  5.100000      0.7500000 0.3050000
##   Rev_Growth Net_Profit_Margin
## 1   23.49000          13.51667
## 2   10.16455          20.17273
## 3    7.01000           6.65000
```

#Count companies in each cluster

```
table(k3$cluster)
```

```
##
##  1  2  3
##  6 11  4
```

K = 4

Calculate average values for each cluster

```
aggregate(Pharma_numeric,
          by = list(Cluster = k4$cluster),
          FUN = mean)
```

```
##   Cluster Market_Cap      Beta PE_Ratio     ROE       ROA Asset_Turnover
## 1       1   55.81000 0.4137500 20.28750 28.7375 12.687500      0.7375000
## 2       2    9.23500 0.6483333 19.43333 17.3000  5.983333      0.4833333
## 3       3  157.01750 0.4800000 22.22500 44.4250 17.700000      0.9500000
## 4       4   26.90667 0.6400000 55.63333 10.1000  4.200000      0.7000000
##    Leverage Rev_Growth Net_Profit_Margin
## 1 0.3712500   5.591250         19.350000
## 2 1.2500000  23.490000         13.516667
## 3 0.2200000  18.532500         19.575000
## 4 0.3166667   6.996667          5.133333
```

Count companies

```r
table(k4$cluster)
```

```
##
## 1 2 3 4
## 8 6 4 3
```

Clustering Method Justification

I selected the nine numerical variables for clustering and excluded the categorical variables. I standardized the variables using the `scale()` function because they have different measurement scales. This gives each variable equal weight in the clustering process.

I used the K-Means algorithm with `nstart = 25` to obtain reliable clustering results. I tested 2, 3, and 4 clusters to compare different groupings and identify the most suitable number of clusters.

#b. Interpret the clusters with respect to the numerical variables used in forming the clusters # Calculate cluster averages for K = 3

```r
cluster_summary <- aggregate(
  Pharma_numeric,
  by = list(Cluster = k3$cluster),
  FUN = mean
)

round(cluster_summary, 2)
```

```
##   Cluster Market_Cap Beta PE_Ratio  ROE   ROA Asset_Turnover Leverage
## 1       1       9.23 0.65    19.43 17.3  5.98           0.48     1.25
## 2       2      97.11 0.43    20.95 35.7 14.95           0.80     0.33
## 3       3      21.75 0.60    46.90 11.3  5.10           0.75     0.30
##   Rev_Growth Net_Profit_Margin
## 1      23.49             13.52
## 2      10.16             20.17
## 3       7.01              6.65
```

The cluster averages help us understand how the companies differ in terms of their financial performance, profitability, growth, and risk.

#c. Is there a pattern in the clusters with respect to the numerical variables (10 to 12)? (those not used in forming the clusters)

```r
#Recreate the cluster vector

set.seed(123)

k3 <- kmeans(Pharma_scaled, centers = 3, nstart = 25)

#Add the cluster label to the pharma
```

```r
Pharma$Cluster_k3 <- k3$cluster

length(Pharma$Cluster_k3)
```

```
## [1] 21
```

```r
#Check the pattern in categorical variables
table(Pharma$Cluster_k3, Pharma$Median_Recommendation)
```

```
##
##     Hold Moderate Buy Moderate Sell Strong Buy
##   1    2            1             0          1
##   2    6            3             2          0
##   3    1            3             2          0
```

```r
table(Pharma$Cluster_k3, Pharma$Location)
```

```
##
##     CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
##   1      1      0       1       0           0  1  1
##   2      0      0       0       0           1  2  8
##   3      0      1       0       1           0  0  4
```

```r
table(Pharma$Cluster_k3, Pharma$Exchange)
```

```
##
##     AMEX NASDAQ NYSE
##   1    0      0    4
##   2    0      0   11
##   3    1      1    4
```

Compare the numerical variables 10–12

```r
cluster_summary_10_12 <- aggregate(
  Pharma[, 12:14],
  by = list(Cluster = Pharma$Cluster_k3),
  FUN = mean
)
```

```
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
```

```
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
## Warning in mean.default(X[[i]], ...): argument is not numeric or logical:
## returning NA
```

```
round(cluster_summary_10_12, 2)
```

```
##   Cluster Median_Recommendation Location Exchange
## 1       1                    NA       NA       NA
## 2       2                    NA       NA       NA
## 3       3                    NA       NA       NA
```

On comparison, three clusters using variables 10–12, which were not used to create the clusters. I also checked the companies' recommendations, locations, and stock exchanges to see if there were any similarities or differences between the groups.

#Cluster 1 has companies with different recommendations, including Strong Buy, Moderate Buy, and Hold.

#Cluster 2 has the highest number of companies, with most receiving a Hold recommendation.Also companies in Cluster 2 are based in the US and listed on the NYSE.

#Cluster 3 mainly has Moderate Buy and Moderate Sell recommendations.Cluster 3 have and Cluster 1 and ore varied locations and exchange listings.

d. Provide an appropriate name for each cluster using any or all of the variables in the dataset.

Cluster 1 – Mixed Recommendation Companies. They have different recommendations, such as Hold, Moderate Buy, and Strong Buy. They are also located in different countries.

Cluster 2 – Large US-Based Companies. Most companies in this group are from the US and listed on the NYSE. Most of them received a Hold recommendation.

Cluster 3 – International Growth Companies. This group includes companies from France, Ireland, and the US. They are listed on different stock exchanges and have different recommendations.

Brief Clarification:

I named each cluster based on its main characteristics. Cluster 2 mostly includes US-based companies listed on the NYSE, while Clusters 1 and 3 have a mix of locations and recommendations, making them more diverse groups.